INF 1004: Mathematics 2

Tutorial : Matrix Algebra & Linear Systems

Trimester 2, 2026

Singapore Institute of Technology

1. Matrix Arithmetic & Shape Constraints

Let the following matrices be defined:

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 6 \end{pmatrix}, B = \begin{pmatrix} -3 & 5 \\ 0 & 9 \end{pmatrix}, C = \begin{pmatrix} 7 & 0 & 4 \\ 1 & 0 & -6 \end{pmatrix}$$

Compute, if mathematically possible:

- (a) $A + B$ and $B + A$
- (b) $A + C$
- (c) $(C^T)^T$

2. Parameter Identification

Find the values of a, b, c, and d to satisfy the equation:

$$\begin{pmatrix} a & 4 & 6 \\ b & 1 & 3 \end{pmatrix} + \begin{pmatrix} c & 2 & 1 \\ 2 & 2 & d \end{pmatrix} = \begin{pmatrix} 5 & 6 & 7 \\ 9 & 3 & 4 \end{pmatrix}$$

3. Non-Commutativity in Multiplication

Given the matrices A and B, compute AB and BA:

$$A = \begin{pmatrix} 3 & 2 \\ 0 & -1 \\ 5 & 2 \end{pmatrix}, B = \begin{pmatrix} 1 & 4 & 2 \\ -2 & 3 & 0 \end{pmatrix}$$

Reinforcement Learning (RL) Context

In RL, transitions between states are often modeled as matrix multiplications. The non-commutative nature ($AB \neq BA$) reminds us that the **sequence of actions** fundamentally changes the resulting environment state.

4. Theoretical Foundations: The Inverse

Problem: Prove that if A and B are invertible $n \times n$ matrices:

$$(AB)^{-1} = B^{-1}A^{-1}$$

- **Hint:** Use the definition $(AB)(AB)^{-1} = I$.
- Multiply (AB) by $(B^{-1}A^{-1})$ and use the associative property.

5. Algebraic Manipulation

Solve for the matrix A:

$$[3A^T + \begin{pmatrix} 4 & 0 \\ 1 & 2 \end{pmatrix}]^T = A + \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}^{-1}$$

Strategy
Apply the property $(M+N)^T = M^T + N^T$ and $(M^T)^T = M$.

Problem: Inverse Matrix Calculation

Problem: Let $B = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$.

Using the methods defined in the lecture:

1. Calculate the **Determinant** $|B|$.
2. Find the **Matrix of Minors** and the **Matrix of Cofactors**.
3. Determine the **Adjoint Matrix**, $adj(B)$.
4. Calculate the **Inverse Matrix**, B^{-1}.

7. Advanced Matrix Properties

Skew-Symmetry and Decomposition

1. **Theorem Proof:** Prove that for any square matrix M, if M is skew-symmetric ($M^T = -M$), then all diagonal entries m_{ii} must be zero.
2. **Decomposition Challenge:** Any square matrix A can be expressed as the sum of a symmetric matrix S and a skew-symmetric matrix V:

$$A = S + V, \quad \text{where } S = \frac{1}{2}(A + A^T) \text{ and } V = \frac{1}{2}(A - A^T)$$

Given $A = \begin{pmatrix} 6 & 8 \\ 2 & 10 \end{pmatrix}$, find S and V and verify if they are symmetric and skew symmetric respectively.